AI-Fit Dual-Phase Nutrition & Workout System



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Submitted to
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DECLARATION

We hereby declare that this project report is based on our original work except for citations and quotations which have been duly acknowledged. We also declare that it has not been previously and concurrently submitted for any other degree or award at Khwaja Fareed University of engineering & Information Technology or other institutions.

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APPROVAL FOR SUBMISSION

I certify that this project report entitled *AI-Fit Dual-Phase Nutrition & Workout System* was prepared by *Muhammad Abuzar* and *Muhammad Ahmed* has met the required standard for submission in partial fulfilment of the requirements for the award of Bachelor of Science in Data Science/Artificial Intelligence at Khwaja Fareed University of Engineering & Information Technology.

Approved by:

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Supervisor: Mr Adeel Abid

Date : _____

MEETING LOG

Meeting No.	Date	Outcome	Signature of Supervisor
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3	17-02-2025	Requirements gathering completed	
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5	24-02-2025	AI algorithms selected	
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ABSTRACT

The AI-Fit Dual-Phase Nutrition and Workout System is a smart application that helps people with two different fitness goals. Some people want to gain fat to build muscles. Some people want to lose fat to reduce weight. This system handles both situations. Most fitness apps give the same plan to everyone. That does not work because every person is different. Our system uses artificial intelligence to create personalized plans. It asks the user for details like age, weight, height, and activity level. Then it calculates how much protein, carbs, and fats the user needs. The system also gives meal suggestions. For example, if you want to gain muscle, it will suggest high protein meals. If you want to lose weight, it will suggest lower calorie meals. It also gives workout plans that match your diet plan. Users can track their progress in the system. They can log their weight and see how they are improving. If the user stops making progress, the system automatically changes the plan.

The system also has a feature called dual-phase switching. This means if a user finishes their muscle building phase, the system will automatically suggest switching to the fat loss phase. We have also taken care of data privacy. User information is encrypted and stored securely. Our goal is to provide a system that is affordable, personalized, and easy to use. This project shows how artificial intelligence can be used in health and fitness.

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CHAPTER 1

INTRODUCTION

1.1. Background

These days fitness has become a very important part of people's lives. More and more people are realizing that they need to take care of their health. Some people want to lose weight, some want to build muscles, and some just want to stay fit and healthy. But there is one big problem that almost everyone faces. Hiring a professional personal trainer is very expensive. A single session can cost anywhere from fifty to one hundred dollars. If someone wants to train three times a week with a trainer, the monthly cost goes into thousands of dollars. Not everyone can afford that much money. So, the question is what do people do when they cannot afford a trainer. Many people turn to fitness apps that are available on their mobile phones. These apps are cheap and some of them are even free. But these apps also have their own set of problems. The first problem is that these apps are not truly personalized. You select a goal like weight loss or muscle gain and the app gives you a standard plan that is almost the same for everyone. The app does not look at your personal strength, your limitations, or your progress over time. The second problem is that when people workout alone, they do not know if they are doing the exercises correctly. You can watch a YouTube video to learn how to do a squat but no one tells you whether your back is straight or your knees are in the right position. When you do exercises with bad form, you do not get the full benefit and more importantly you can get injured. This is a serious risk that many people do not think about. The third problem is motivation. When you work out with a trainer, that person encourages you, pushes you to do better, and holds you accountable. But when you are using an app alone, no one is forcing you to complete your workout. One day you feel tired and tell yourself that you will do it tomorrow. That tomorrow never comes. This is why so many people start fitness journeys but very few actually stick with them for a long time.

This is where the idea of an AI Fitness Coach comes in. This system is designed to give users personalized workout plans, track their progress, give them real time feedback on their form, and keep them motivated. It cannot fully replace a human trainer but for people who cannot afford a trainer, it is a very good alternative. The system is available twenty-four hours a day, seven days a week. It does not get tired and it does not charge high fees.

1.2. Problem Statement

Even though there are many fitness applications available in the market, several major problems still remain unsolved. The first problem is the lack of true personalization. Most applications ask the user to select a goal and then provide a standard plan that was created in advance. This plan does not change based on how the user is progressing. It does not consider whether the user is a beginner or advanced. It does not learn from the user's past performance. The user is basically getting the same experience as everyone else. The second problem is incorrect exercise form. When users follow written instructions or watch video demonstrations, they often misunderstand the important details of proper form. They may round their back during a deadlift. They may let their knees cave inward during a squat. They may use momentum instead of muscle control during bicep curls. Without someone watching them and correcting them in real time, these mistakes become habits. Over time these bad habits lead to injuries that could have been easily prevented. The third problem is the lack of continuous motivation. A human personal trainer shows up at scheduled times, expects the client to be ready, and provides encouragement throughout the session. A fitness application has no expectations and provides no social pressure. When a user feels tired or unmotivated, skipping a workout has no immediate consequence. Over time skipped workouts become more frequent and eventually the user stops using the application entirely. The fourth problem is the high cost of professional training. As mentioned earlier, hiring a personal trainer is very expensive. For a student or a young professional just starting their career, spending hundreds of dollars each month on a trainer is simply not possible. There is a clear gap between expensive human trainers and cheap but ineffective fitness apps. The fifth problem is poor progress tracking. Most applications let users log their workouts but they do not provide any real analysis. They do not tell the user whether they are improving fast enough or too slowly. They do not suggest when to increase weights or add more repetitions. The user has to figure all of this out on their own.

1.3. Objectives

The following objectives have been set for the AI Fitness Coach project. The first objective is to design and build an AI powered system that gives personalized workout guidance to users based on their individual fitness levels, goals, and preferences. The second objective is to create a proper user management system where people can register for an account, login securely, and update their personal information including age, weight, height,

fitness level, and specific goals like weight loss or muscle gain. The third objective is to build a personalized workout plan generator that uses machine learning to recommend exercises, sets, repetitions, and rest periods that are right for each user. The system will look at the user's current fitness level and their goals to produce plans that are challenging but doable. The fourth objective is to add exercise tracking features including automatic repetition counting, set tracking, and workout duration monitoring. If we can get the computer vision part working, the system will also detect body posture and give feedback on exercise form. The fifth objective is to provide clear step by step instructions for each exercise. Before a user starts an exercise, the system will show them exactly how to do it, which muscles should be working, and what common mistakes to avoid. The sixth objective is to store all user data including workout history, performance metrics, and progress over time in a proper database. This historical data will be used to show progress charts and to make future workout recommendations better. The seventh objective is to apply artificial intelligence and machine learning techniques to solve a real-world problem in fitness and health. This is not just theoretical research. The goal is to produce a working system that actually helps people. The eighth objective is to test the system with real users and measure how well it works in terms of user engagement, satisfaction, and fitness improvements.

1.4. Project Scope

The scope of this project includes the design and development of an AI based fitness coaching system that provides personalized workout guidance, tracks user performance, and offers intelligent feedback. The system is focused on making personal fitness training more accessible, more affordable, and more effective through the use of artificial intelligence. The features that are included in the scope are user registration and login, collection of personal data like age, weight, height, fitness level and goals, AI based workout plan recommendations, customized exercise routines, repetition counting, set tracking, workout duration monitoring, step by step exercise instructions, start and stop functionality for workout sessions, recording of completed workouts, database storage for user and workout data, and progress tracking over time. The features that are not included in the scope are diet and nutrition planning which would need a separate system, integration with wearable devices like smartwatches, real time video streaming to human trainers, medical diagnosis or rehabilitation guidance, social features like friend challenges and leader boards, and native mobile apps for iOS and Android beyond responsive web design. The target users for this system are students and young professionals

who want to work out at home but cannot afford expensive personal trainers, home workout users who need guidance and motivation, and fitness enthusiasts who want to track their progress more systematically.

This project is classified as an Applied Project. This means we are focused on solving a real-world problem by using existing theories, tools, and technologies. We are not trying to invent new algorithms or create new theories. Instead, we are taking what already exists and applying it to make something useful. We will implement machine learning for workout personalization. We will apply computer vision for exercise monitoring. We will build a working prototype that real users can try. We will solve a practical problem in the fitness domain.

1.5. Advantages of the System

The AI Fitness Coach system has several advantages over traditional fitness apps and even over human personal trainers in some ways. The biggest advantage is accessibility. A human trainer is only available during scheduled sessions. An AI coach is available all the time, any time of day or night. If someone wants to work out late at night or early in the morning, they can do that without waiting for an appointment. The second advantage is affordability. A human personal trainer costs hundreds or even thousands of dollars per month. An AI powered system can be offered at a much lower price. With a freemium model where basic features are free and premium features cost a small monthly fee, more people can access quality fitness guidance. The third advantage is personalization. Unlike generic apps that give everyone the same plan, the AI system learns from each user's performance. If a user is improving quickly, the system will give them more challenging workouts. If a user is struggling with certain exercises, the system will offer modifications or alternatives. The fourth advantage is real time form feedback. When a user does an exercise incorrectly, the system can detect this through the camera and give immediate correction. This helps prevent injuries and makes sure the user gets the full benefit of each exercise. The fifth advantage is consistency and accountability. The system tracks every workout and provides reminders and encouragement. Users can see their progress through charts and graphs, which helps keep them motivated to continue.

1.6. Relevance to the Study Program

This project is very relevant to the Bachelor of Science in Artificial Intelligence program for several reasons. First, we are implementing machine learning algorithms for personalization. This is a core topic in the AI curriculum. Second, we are using computer vision techniques for pose detection and repetition counting. This shows how image and video processing can be applied in real life. Third, we are following a proper software development process including requirement engineering, design, implementation, testing, and deployment. Fourth, we are solving a real-world problem in health and fitness, which shows how AI can create value for society. Fifth, we are working with databases to store and manage user data. Sixth, we will end up with a working prototype that we can demonstrate and evaluate.

1.7. Chapter Summary

This chapter introduced the AI Fitness Coach project. We talked about why AI based fitness coaching is needed when human trainers are expensive and generic apps are not effective. We discussed five main problems including lack of personalization, incorrect exercise form, lack of motivation, high cost, and poor progress tracking. We listed eight objectives for the project. We defined the scope by saying what features are included like user management, personalized plans, and exercise tracking, and what features are excluded like diet planning and wearable integration. We talked about the advantages of the system including accessibility, affordability, personalization, real time feedback, and accountability. Finally, we explained why this project is relevant to our degree program. The next chapters will cover existing systems, requirement engineering, system design, and database architecture.

CHAPTER 2

EXISTING SYSTEMS

2.1. Existing Systems

Before building our own system, we looked at what is already available in the market. We wanted to understand what features existing applications offer and where they fall short. We looked at several popular fitness applications including MyFitnessPal, Frenetic, Fitbod, Kaia Health, Nike Training Club, and Future. Freeletics is one of the older AI driven fitness apps. It has been around since 2013 and has been downloaded over fifty million times. Freeletics uses an algorithm called the Freeletics Coach to generate bodyweight workout plans. After each workout, the user rates how difficult it was, and the algorithm adjusts future workouts based on that rating. Freeletics also provides video demonstrations of each exercise. But there are some things Freeletics does not do. It does not have computer vision for form correction. It does not count repetitions automatically. The user has to count their own reps and enter them manually. Also, Freeletics is mostly focused on bodyweight exercises and does not give good guidance for people who have access to gym equipment. Fitbod is another AI driven workout app that has become popular among people who do strength training. Fitbod uses machine learning to generate workouts based on the equipment the user has available, how often they want to train, and their fitness goals. The app also tracks which muscles have recovered and suggests exercises that target the muscles that are ready to work. Fitbod adapts the number of sets and repetitions based on how the user performed in previous workouts. But like Freeletics, Fitbod does not have computer vision for form correction. The user has to enter their repetitions and weights manually. The app cannot see whether the user is doing the exercises correctly.

Kaia Health is interesting because it does use computer vision to analyse user posture during exercises. The app gives real time feedback on form and suggests corrections. Kaia Health has actually been studied in clinical trials and has been shown to help reduce back pain and improve physical function. However, Kaia Health is focused mainly on physical therapy and rehabilitation rather than general fitness training. The exercise library is limited compared to regular fitness apps. Also, Kaia Health does not generate personalized workout plans based on user goals and performance. It is more about doing specific therapeutic exercises correctly.

Nike Training Club is one of the most popular fitness apps and it is completely free. It offers hundreds of guided workouts led by professional trainers. The app includes video demonstrations, audio coaching, and some basic personalization based on user preferences. But Nike Training Club does not use machine learning to adapt workouts based on user performance. The personalization is very basic. You tell the app what your goals are and it suggests some workouts, but it does not learn from your progress over time. It also has no computer vision features. Future is a different kind of app. It pairs users with real human personal trainers who design custom workouts and provide feedback. Future combines the convenience of an app with the expertise of a real person. But Future is very expensive at about one hundred fifty dollars per month. That is cheaper than in person training but still too expensive for many people. Also, the quality of coaching depends entirely on which trainer you get assigned.

2.2. Drawbacks in Existing Systems

After looking carefully at these existing systems, we found several common problems. The biggest problem is that no single app does everything well. Apps that provide good AI driven personalized workout plans like Freeletics and Fitbod do not have computer vision for form correction. Apps that have computer vision for form correction like Kaia Health do not provide good personalized workout plan generation. Users who want both features have to use two different apps, which is inconvenient. The second problem is manual data entry. In most existing apps, users have to count their own repetitions and enter the numbers after each set. This interrupts the workout and takes focus away from doing the exercise properly. It also relies on the user being honest and accurate. People might forget to log a set or they might lose count. Some people might even inflate their numbers to feel better about themselves. An ideal system would count repetitions automatically without the user having to do anything. The third problem is the lack of real time feedback.

Most apps show a video demonstration before the exercise starts, but once the user begins moving, the app stops helping. If the user's form gets worse halfway through a set, the app cannot detect this and cannot give a correction. By the time the user finishes the set, they have already done several repetitions with bad form. The fourth problem is that even the so-called personalized apps are not truly using machine learning the right way. Many of them use simple rules. For example, if the user says a workout was easy, increase the difficulty by some percentage. This is not really machine learning. It is just a rule. True machine learning would

look at patterns across many users and identify complex relationships that simple rules cannot capture. The fifth problem is cost. Apps that do provide human coaching like Future are effective but expensive. The whole point of building an AI Fitness Coach is to provide intelligent guidance at a much lower cost than a human trainer. But most low-cost apps lack the intelligence and feedback features to actually replace a trainer.

2.3. Examples of Existing Systems

Let me give three specific examples in more detail. The first example is Freeletics. When a user signs up for Freeletics, they answer some questions about their fitness level and goals. The app then generates a bodyweight workout plan. After each workout, the user rates the difficulty from one to five. Based on that rating, the app adjusts future workouts. If the user rates a workout as easy, the next one will be harder. If they rate it as very hard, the next one will be easier. This is a simple but effective system. However, Freeletics does not track your form at all. You could be doing every exercise completely wrong and the app would never know Also, you have to count your own repetitions.

The second example is Fitbod. Fitbod asks about your available equipment, how many days per week you want to train, and what your goals are. It then generates workouts that include specific exercises, sets, and repetitions. The app tracks your muscle recovery and makes sure you do not work the same muscles two days in a row. As you log your workouts, Fitbod learns how much weight you can handle for each exercise and adjusts future recommendations. But again, no form feedback. You could be lifting the right weight and doing the right number of reps but with terrible form, and Fitbod would have no idea. The third example is Kaia Health. This app uses your phone camera to watch you while you exercise. It detects your body position using computer vision and tells you if you are doing something wrong. For example, if you are doing a squat and your back is rounding, the app might say straighten your back. This is very useful for people doing physical therapy exercises. But Kaia Health does not generate personalized workout plans. It has a fixed set of therapeutic exercises. It is not designed for someone who wants to build muscle or lose weight through general fitness training.

2.4. Need to Replace Existing Systems

Looking at all these problems, it is clear that there is room for a better system. Users deserve an app that combines AI driven personalized workout plans with computer vision-

based form correction and automatic repetition counting. No existing app does all three of these things together. Users also deserve real time feedback during the exercise, not just a video at the beginning.

A system that watches the user through the camera and gives corrections as they happen would help people learn proper form much faster and would reduce the risk of injury. Users deserve automatic tracking that does not require manual entry. They should be able to focus on their work out without having to count reps or log sets. The system should do all of that automatically. Users deserve true machine learning personalization that actually learns from their data over time. Not just simple rules, but real algorithms that identify patterns and make predictions. And finally, users deserve all of this at a price that is affordable for students and young professionals. That is exactly what we are trying to build with the AI Fitness Coach.

2.5. Chapter Summary

This chapter looked at existing fitness applications and identified their limitations. We examined Freeletics, Fitbod, Kaia Health, Nike Training Club, and Future. The main problems we found were that no single app provides both good personalization and good form feedback, most apps require manual repetition counting, real time feedback is missing, the so-called personalization is often just rule based, and good solutions are too expensive. We discussed three examples in detail to show what is currently available and what is missing. Finally, we explained why there is a need for a new system that combines all the important features in one affordable package.

CHAPTER 3

REQUIREMENT ENGINEERING

3.1. Proposed System

The AI Fitness Coach system we are proposing is designed to fix all the problems we found in existing systems. Unlike other apps that only do one thing well, our system will combine AI driven personalized workout plans with computer vision-based form correction and automatic repetition counting all in one platform. Users will not need to use one app for planning their workouts and another app for checking their form. Everything will be in one place. The system will work as a web-based application. Users can access it from their computer if it has a camera, or from their mobile phone. To get started, a user will create an account and fill out a profile. The profile will ask for basic information like age, weight, height, fitness level, and fitness goals. Based on this information, the system will generate a personalized workout plan. The plan will include specific exercises, how many sets to do, how many repetitions in each set, how long to rest between sets, and how hard to push. The thing that makes our system different is the computer vision part. Using libraries like OpenCV and Media Pipe, the system will use the device camera to watch the user while they exercise. It will detect the user's body position and compare it to the correct form for that exercise. If the user's form is wrong, the system will give immediate feedback. For example, if the user is doing a squat and their back is rounding, the system might say keep your back straight.

This feedback happens in real time while the user is still doing the exercise, so they can correct themselves immediately. The system will also count repetitions automatically. The user does not have to keep track of how many reps they have done. The camera will detect each repetition and count it. This lets the user focus completely on doing the exercise correctly rather than on counting numbers in their head. The machine learning part of the system will look at how the user performs over time. If a user consistently completes all their reps with good form, the system will make their workouts more challenging. It might increase the number of reps or add more sets. If a user is struggling with a particular exercise, the system might suggest an easier variation or a different exercise that works the same muscles.

3.2. Understanding the System

To build a good system, we first need to understand how people will use it, who has an interest in it, what domain it belongs to, and what specific needs it must meet.

3.2.1. User Involvement

Users will interact with the system at several different levels. First, they will create their profile by providing their personal information and fitness goals. This is a one-time thing but they can update it later if something changes. During each workout session, users will position themselves in front of their device camera. The system will show them what exercise to do next, how many reps to aim for, and how many sets are remaining. As they perform the exercise, the system will count reps and give form feedback. After each set, the system might ask them to rate how hard the set was. After the workout is finished, users can see a summary of how they did include total reps completed, sets completed, and any form issues that were detected. Over time, users can look at progress charts to see how they have improved. They can see trends in their strength, their consistency, and their form quality. Users can also give feedback on the workout plans. If an exercise feels too hard or too easy, they can tell the system and it will adjust future recommendations.

3.2.2. Stakeholders

There are several groups of people who have a stake in this project. The most important group is the end users. These are the people who will actually use the system to help them with their fitness goals. They care about whether the system works well, whether it is easy to use, whether their data is private, and whether they actually see results.

The development team is the next group. This includes the two students Muhammad Abuzar and Muhammad Ahmed and the project supervisor Mr. Adeel Abid. They care about completing the project successfully, learning new skills, and creating something that works. The Department of Data Science and AI at Khwaja Fareed University also has an interest because this project reflects on the quality of their program and their students. In the future, other stakeholders could include fitness professionals who might use the system as a tool for their clients, and gym owners who might offer the system as a service to their members.

3.2.3. Domain

This system sits at the intersection of three different domains. The first is artificial intelligence and machine learning. We are using algorithms to personalize

workouts and learn from user data. The second is computer vision. We are using pose detection to watch users and give form feedback. The third is health and fitness technology. We are applying exercise science principles to make sure our recommendations are safe and effective.

3.2.4. Needs of System

Based on our research and discussions with potential users, we have identified ten key needs that the system must satisfy. These are listed in the table below.

Table 3.1 User Needs of System

SR	Need	Need ID
1	The system must let users create accounts and login securely	N-01
2	The system must collect user information like age, weight, height, fitness level, and goals	N-02
3	The system must generate workout plans that are personalized to each user	N-03
4	The system must show clear instructions for each exercise	N-04
5	The system must count repetitions automatically without the user doing anything	N-05
6	The system must track sets and workout duration	N-06
7	The system must give real time feedback on exercise form using the camera	N-07
8	The system must save user workout history for progress tracking	N-08
9	The system must work on both desktop computers and mobile devices	N-09
10	The system must be affordable compared to human trainers	N-10

3.3. Requirement Engineering

Requirement engineering is the process of figuring out what the system needs to do and writing it down in a clear way. This section covers the functional requirements, non-functional requirements, and how each user need maps to specific features.

3.3.1. Functional Requirements

Functional requirements describe what the system must actually do. Each one has a unique ID, a name, a description, what inputs it takes, what outputs it produces, what conditions must be true before it runs, and what will be true after it runs.

Table 3.2 Functional Requirement FR-01

Element	Description
Functional Requirement ID	FR-01
Name	User Account Management
Description	The system will allow users to create an account, login, logout, and update their profile information.
Input	For registration name email and password. For login email and password.
Output	Account creation confirmation and a session token for authenticated access.
Precondition	For registration the email must not already be in use. For login the user must have an existing account.
Postcondition	An authenticated user can access personalized features.

Table 3.3 Functional Requirement FR-02

Element	Description
Functional Requirement ID	FR-02
Name	User Profile and Goal Setting
Description	The system will collect user information including age weight height gender fitness level and fitness goals like weight loss muscle gain or endurance improvement.

Input	Age in years, weight in kilograms, height in centimetres, gender, fitness level (beginner, intermediate, advanced), primary goal (weight loss, muscle gain, endurance, general fitness)
Output	User profile stored in the database and baseline fitness metrics like BMI.
Precondition	The user has completed account registration and is logged in.
Postcondition	The user profile is ready for generating personalized workout plans.

Table 3.4 Functional Requirement FR-03

Element	Description
Functional Requirement ID	FR-03
Name	Personalized Workout Plan Generation
Description	The system will use AI to generate workout plans including exercise sets repetitions rest periods and intensity recommendations based on user goals and fitness level.
Input	User profile data and past workout performance if available.
Output	A weekly or daily workout plan with exercise lists, prescribed sets, prescribed reps, rest periods, and intensity guidance.
Precondition	The user profile is complete and goals are set
Postcondition	The workout plan is displayed to the user and ready to start.

Table 3.5 Functional Requirement FR-04

Element	Description
Functional Requirement ID	FR-04
Name	Exercise Execution and Tracking

Description	The system will start and stop workout sessions, show step by step instructions, count repetitions using computer vision, track sets completed, and monitor workout duration.
Input	Camera video feed for repetition counting and form analysis, user commands to start or stop.
Output	Real time repetition count, set completion status, workout duration, and form quality alerts.
Precondition	The user has started a workout session and camera access has been granted.
Postcondition	Completed workout data is saved to user history.

Table 3.6 Functional Requirement FR-05

Element	Description
Functional Requirement ID	FR-05
Name	Real Time Form Feedback
Description	The system will use computer vision to analyse user body posture during exercises and will give immediate feedback when form problems are detected.
Input	Camera video feed and pose landmark coordinates from media Pipe.
Output	Form quality score and specific correction suggestions like straighten your back or lower your knees.
Precondition	The camera is active and the user is performing an exercise that has pose detection configured.
Postcondition	The user receives feedback that can be used to correct form immediately.

Table 3.7 Functional Requirement FR-06

Element	Description
Functional Requirement ID	FR-06
Name	Progress Tracking and Analytics
Description	The system will save completed workouts, store historical performance data, and show visual analytics of progress over time.
Input	Completed workout data including exercises reps sets form quality scores and duration.
Output	Progress charts showing trends in volume, consistency, form quality, and goal achievement.
Precondition	The user has completed at least one workout session.
Postcondition	The user can view their progress and the system uses the data for better personalization.

3.3.2. Non-Functional Requirements

Non-functional requirements describe how well the system performs, not what it does. Here are the non-functional requirements for this project. The system should respond to user commands within one second for basic things like starting a workout. For computer vision processing, it should process at least fifteen frames per second to give smooth real time feedback. The system must keep user data private and secure. Camera feeds should not be stored permanently. User data should not be shared with third parties without permission. The system should be easy to use for anyone with basic computer skills. You should not need to understand AI or computer vision to use it. The system should work on both desktop computers and mobile devices. On devices without cameras, the form feedback features will not work but the rest of the system will still function. The system should be affordable with basic features free and premium features available for a small monthly subscription of no more than ten dollars.

3.3.3. Requirements Baseline

The requirements baseline includes all the functional requirements from FR-01 to FR-06 and all the non-functional requirements listed above. Any changes to these requirements after the baseline are set will need formal approval from the project supervisor. The baseline has been reviewed and agreed to by both student developers Muhammad Abuzar and Muhammad Ahmed and supervisor Mr. Adeel Abid.

3.3.4. Need to Feature Mapping

Each user need from Table 3.1 connects to specific features. Need N-01 for account management is covered by FR-01. Need N-02 for collecting user data is covered by FR-02. Need N-03 for personalized plans is covered by FR-03. Need N-04 for instructions is covered by FR-04. Need N-05 for automatic rep counting is covered by FR-04. Need N-06 for set and duration tracking is covered by FR-04. Need N-07 for real time form feedback is covered by FR-05. Need N-08 for progress tracking is covered by FR-06. Need N-09 for multi device access is covered by the responsive design requirement. Need N-10 for affordability is covered by the freemium business model.

3.4. Gantt Chart

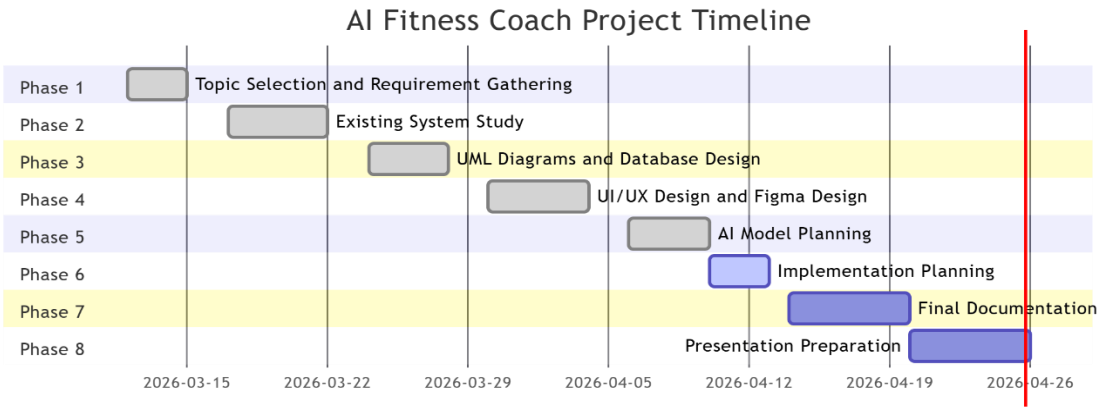


Figure 3.4 Gantt Chart

3.5. Hurdles in Optimizing the Current System

There are some challenges we expect to face while building this system. The first challenge is getting the computer vision pose detection to work accurately in different lighting conditions. Some users might exercise in dim rooms or with bad camera angles, which could

make it harder for the system to see their body clearly. The second challenge is performance. Processing video frames in real time takes a lot of computing power. Not everyone has a high-end computer. We will need to optimize our code to run smoothly on average devices. The third challenge is that people have different body shapes and sizes. Some pose detection models work well for average body types but not as well for people who are very tall, very short, or have different proportions. Loose clothing can also make pose detection harder. The fourth challenge is that some users might be uncomfortable with the camera access, even if we tell them we are not storing the video. We will need to have clear privacy policies and give users the option to turn off camera features if they want. The fifth challenge is the cold start problem. When a new user joins, we have no data about them yet. The system has to rely on the initial profile information to generate the first workout plan. It will get better over time as the user completes more workouts.

3.6. Chapter Summary

This chapter covered the requirement engineering for the AI Fitness Coach project. We described the proposed system as an integrated platform combining personalized workout plans, computer vision form feedback, and automatic rep counting. We explained user involvement, stakeholders, domain, and user needs in a table. We wrote six functional requirements with complete details. We listed non-functional requirements for performance, security, usability, accessibility, and affordability. We established the requirements baseline and showed how each user need maps to specific features. We presented a Gantt chart for the sixteen-week timeline. Finally, we discussed five challenges we expect to face including pose detection accuracy, performance, body diversity, privacy concerns, and the cold start problem.

CHAPTER 4

DESIGN

4.1. Software Process Model

For this project we decided to use the Agile software development model. We chose Agile because our project has several different parts that need to work together including the user interface, the backend server, the computer vision system, and the machine learning models. These parts cannot be built completely separately because what we learn while building one part might change how we build another part. Agile lets us adjust as we go. In the Agile model, we break the work into sprints. Each sprint is two weeks long. At the start of each sprint, we decide what features we will work on. At the end of the sprint, we should have worked software that we can show and test. We then get feedback and use that feedback to plan the next sprint. For example, imagine we spend one sprint working on the computer vision pose detection. We might discover that the pose detection works well in good lighting but fails in low light. In the next sprint we can prioritize fixing that problem. With a non-Agile approach like Waterfall, we would have to go back and change our design documents first, which takes more time.

4.1.1. Benefits of Selected Model

The Agile model gives us several benefits for this project. The first benefit is flexibility. We are trying out different computer vision libraries and machine learning approaches. Some will work well and some will not. Agile lets us change direction quickly when something is not working. The second benefit is that we can deliver working software early. By the end of the second or third sprint, we can have a basic version running where users can register, login, and see a simple workout plan. Even though the computer vision feature might not be ready yet, we can still show progress to our supervisor. The third benefit is that risks get identified early. If we try to implement real time pose detection and find out that our computers are too slow to process fifteen frames per second, we will know this in the first few weeks rather than finding out at the end of the project.

4.1.2. Limitations of Selected Model

Agile also has some limitations. The biggest one is that it requires constant communication among team members. We are two students working on this project. If we do not talk to each other regularly, things can get messy. We have committed to meeting twice a week to discuss progress and plan the next sprint. Another limitation is that documentation can fall behind. In Agile the focus is on working software, not on writing documents. But our university requires a proper project report. We have to make sure we do not ignore the documentation while we are busy coding.

4.2. Design

The design of our system includes the overall methodology, database design, and UML diagrams. We will go through each of these in this section.

4.2.1. Methodology of the Proposed System

Our system follows a five-step methodology. The first step is data collection. We collect user information during registration and we collect workout performance data every time the user exercises. This data is the fuel for our machine learning models.

4.2.1.1 Methodology Diagram

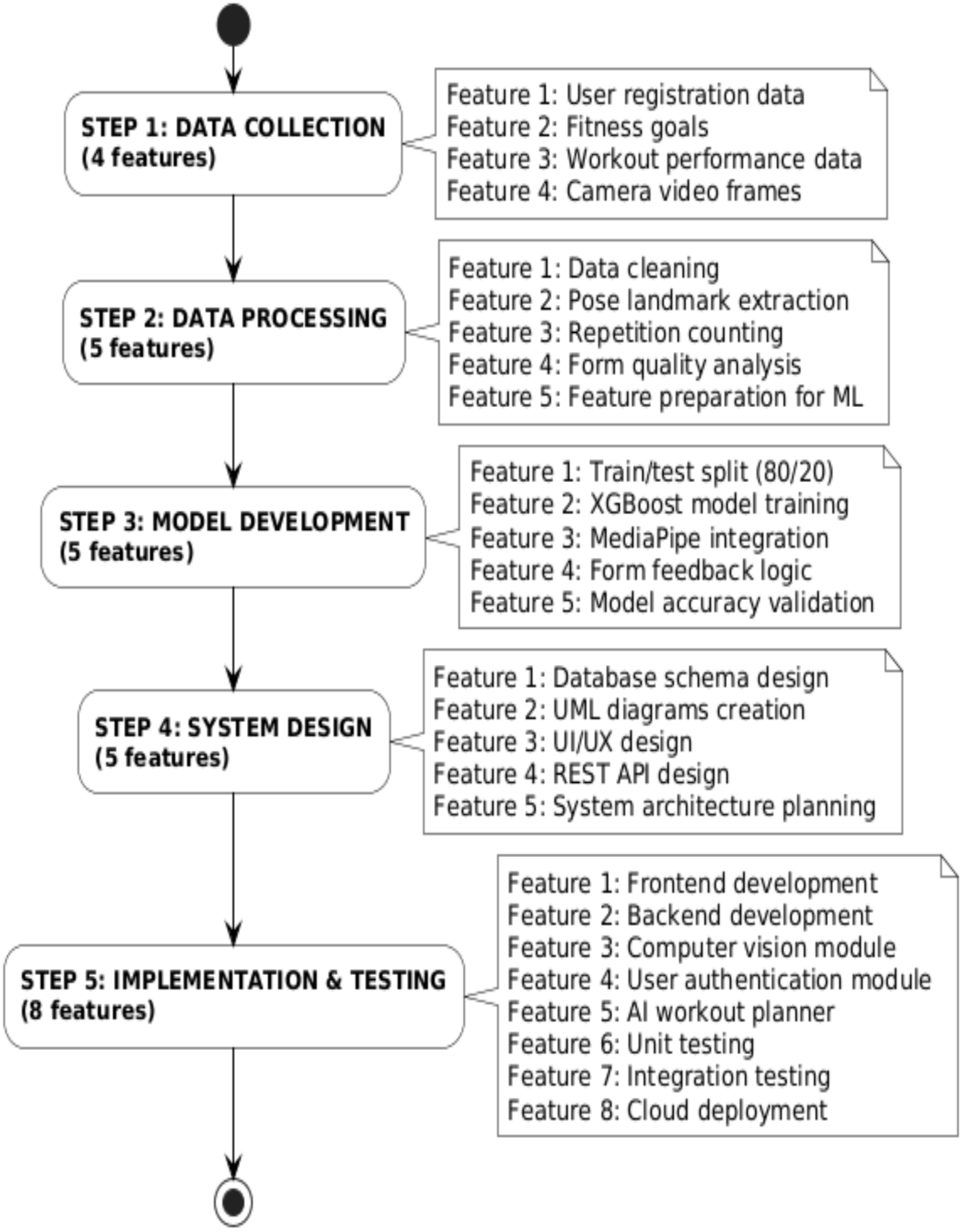


Figure 4.1 Methodology Diagram

4.2.2. Entity Relationship Diagram

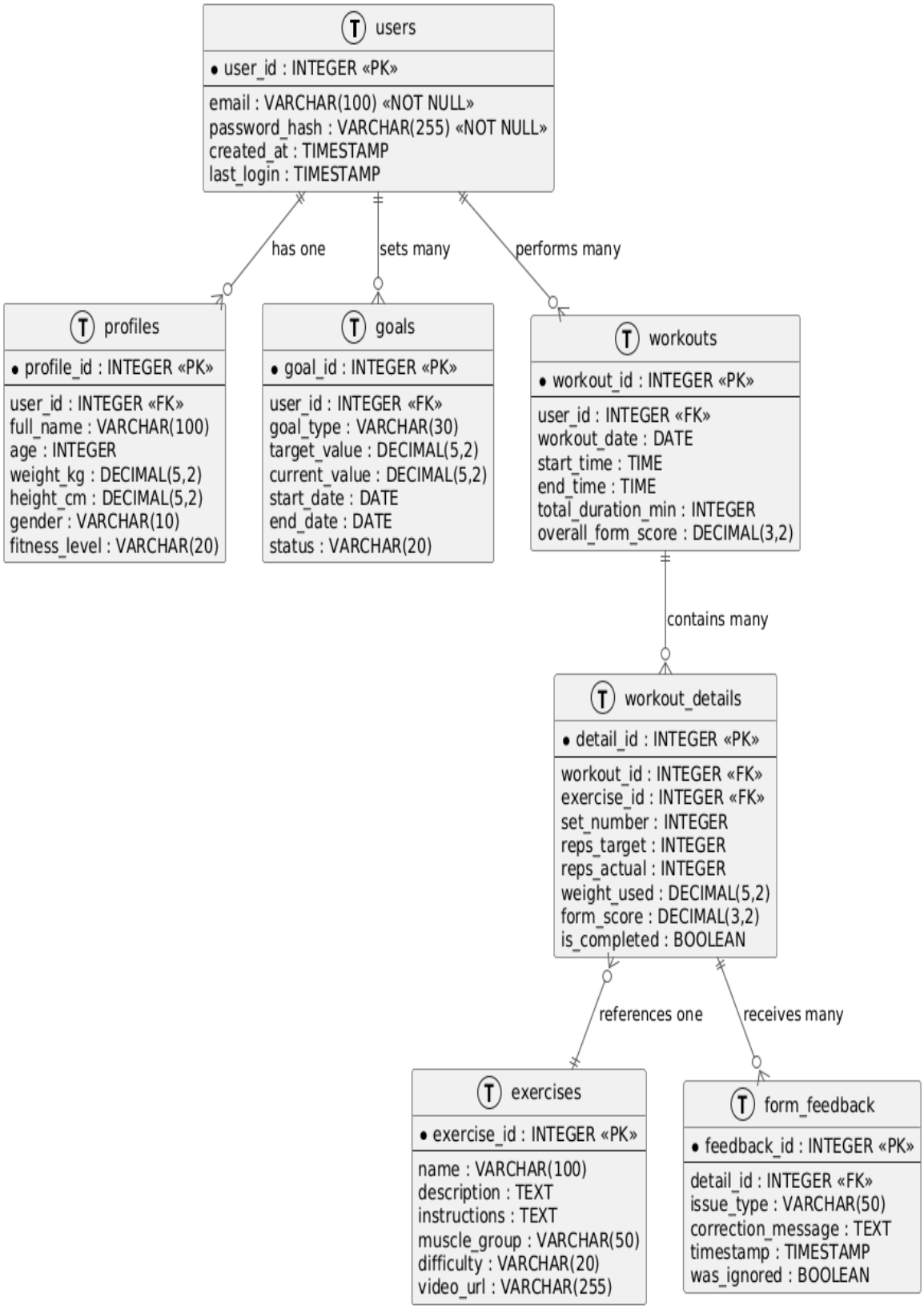


Figure 4.2 ERD Diagram

4.2.3. UML Diagrams

We created several UML diagrams to help us visualize how the system will work. These are standard diagrams used in software engineering to communicate design ideas.

4.2.3.1. Use Case Diagram of the System

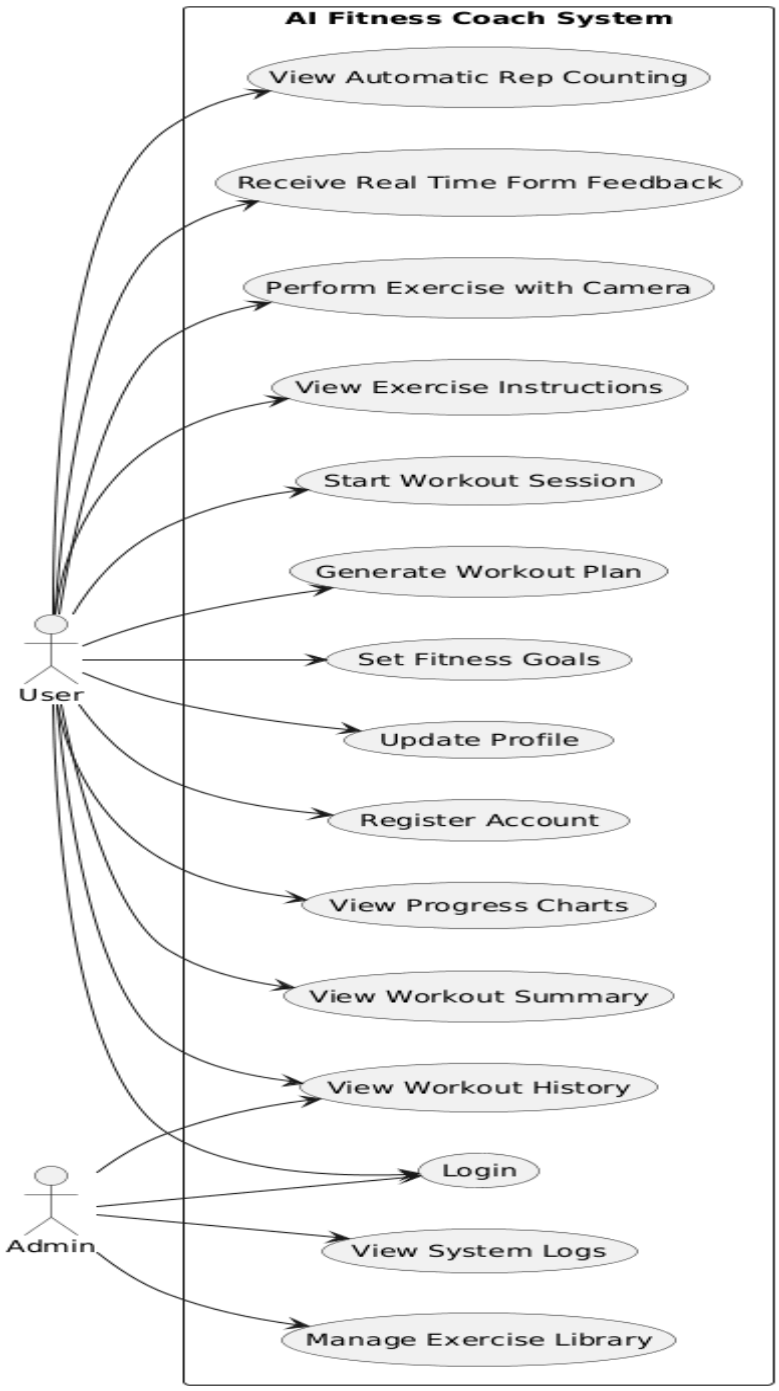


Figure 4.3 Use Case Diagram

4.2.3.2. Class Diagram of the System

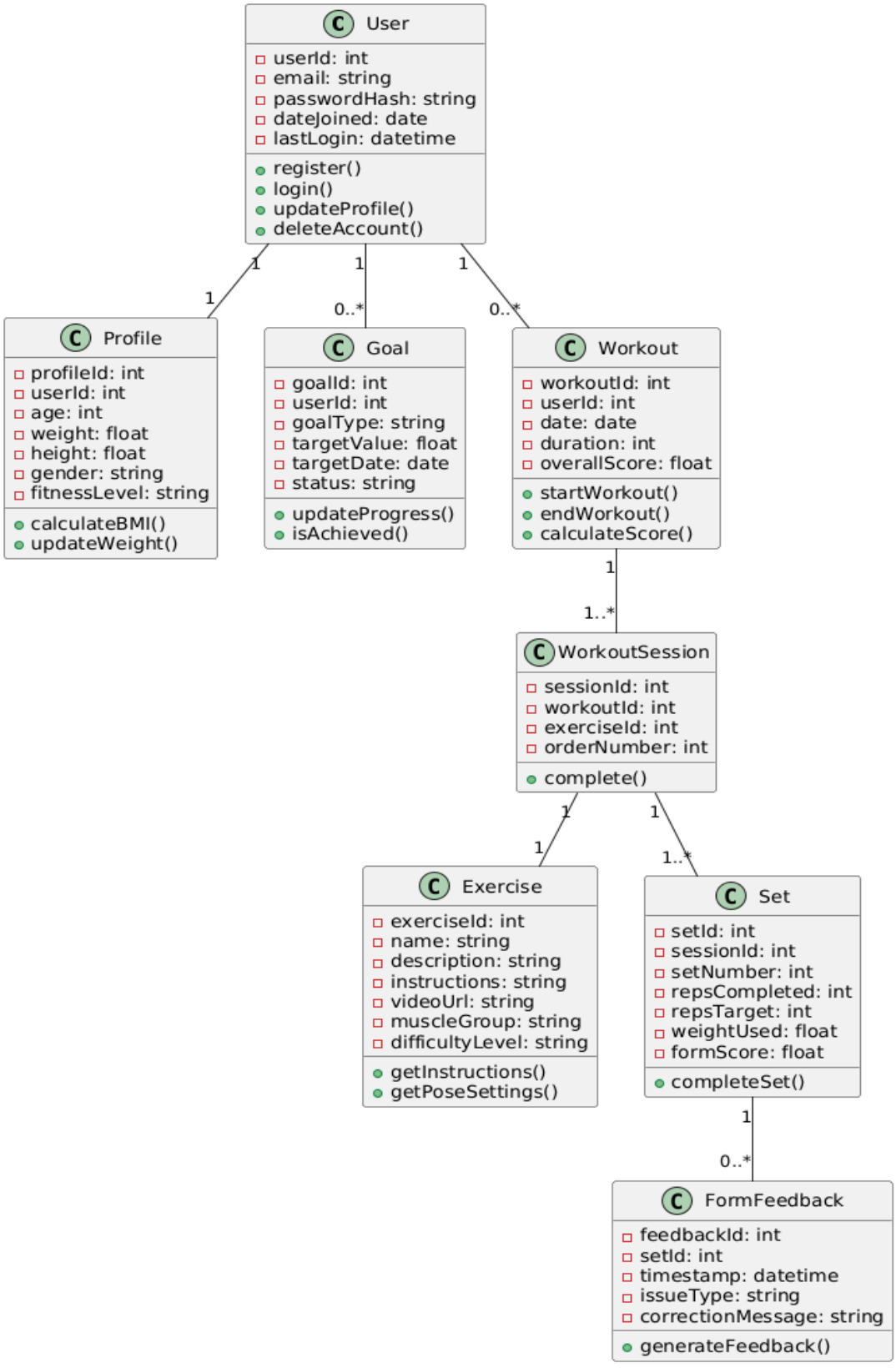


Figure 4.4 Class Diagram

4.2.3.3. Activity Diagram of the System

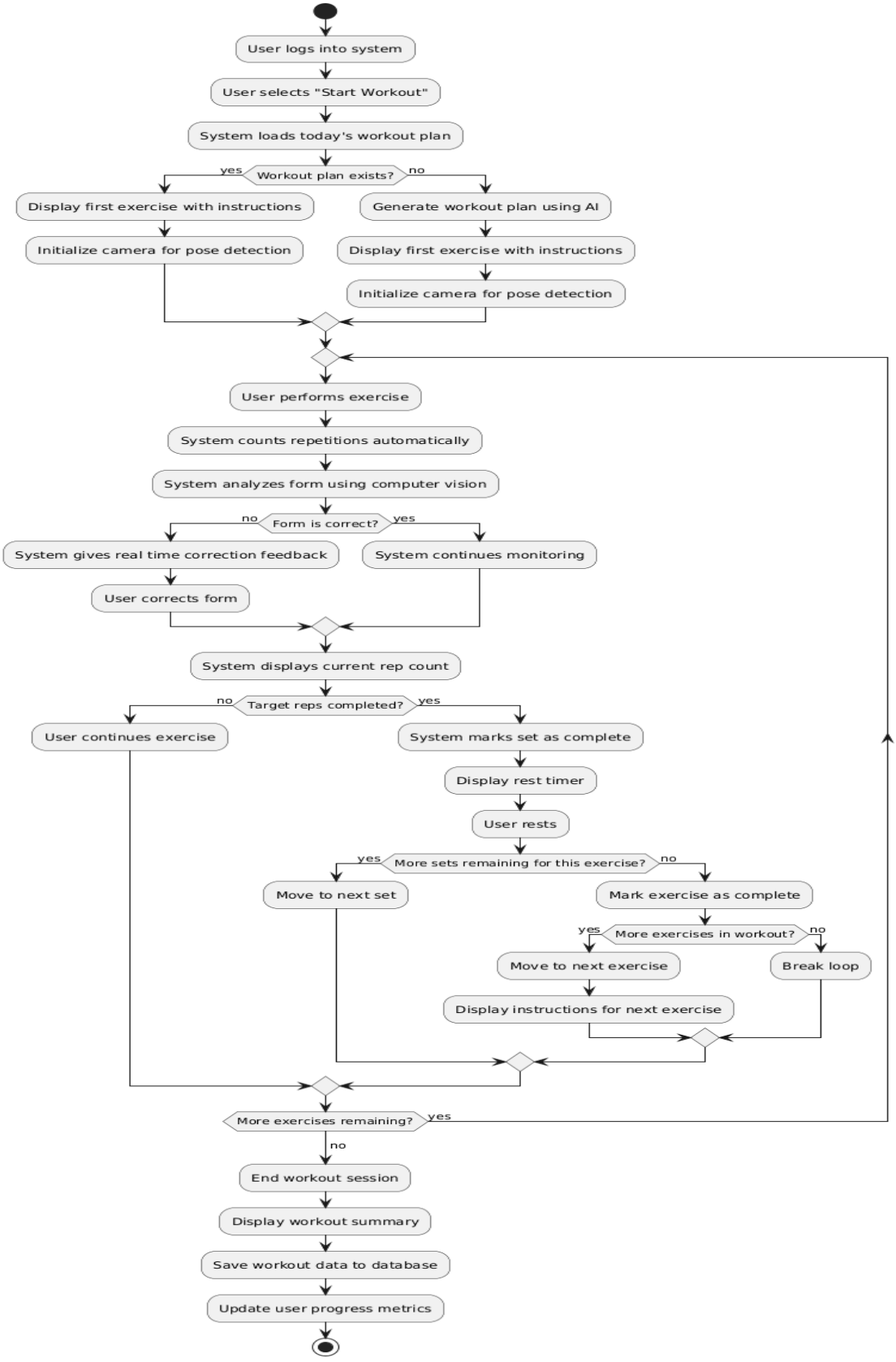


Figure 4.5 Activity Diagram

4.2.3.4. Sequence Diagram of the System

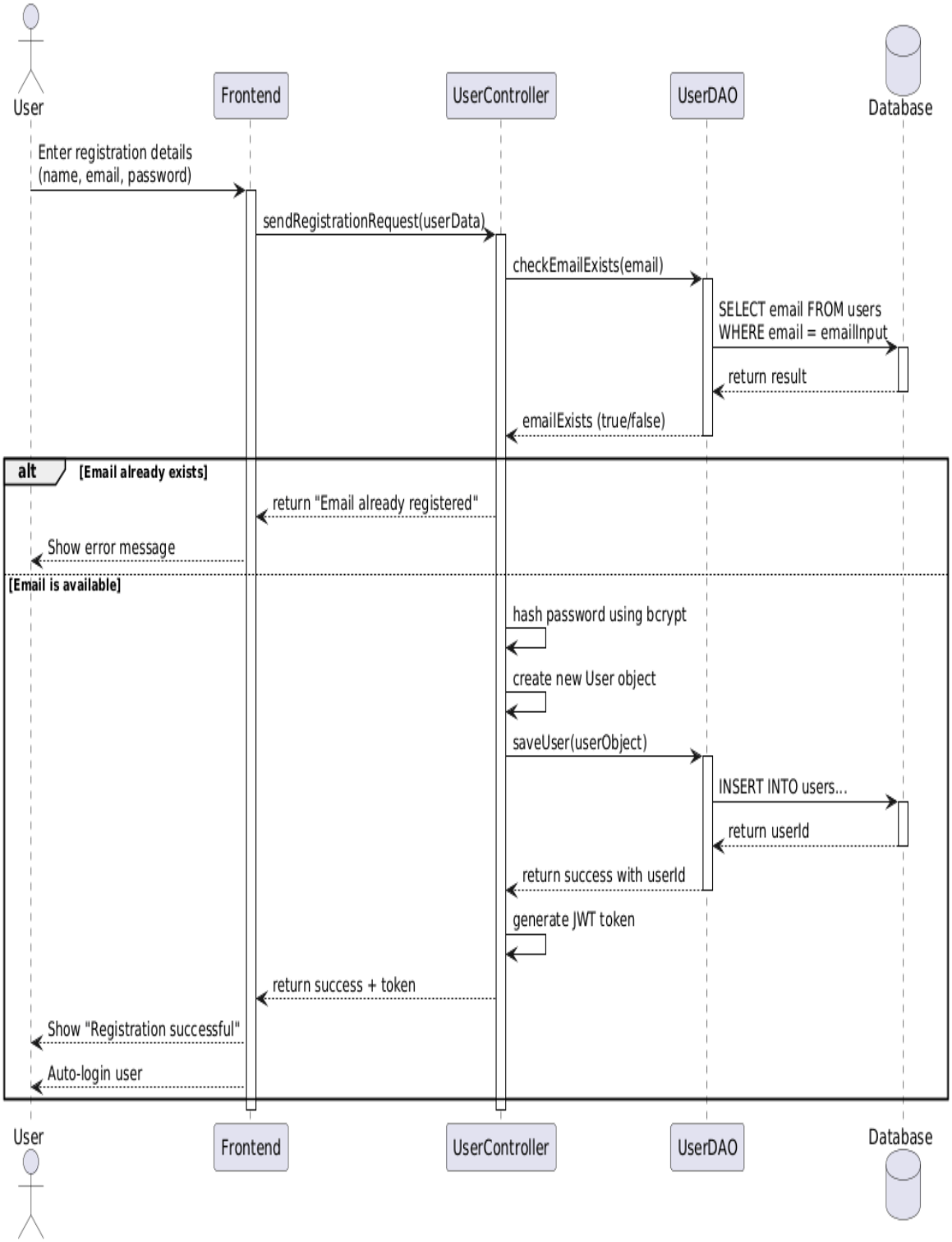


Figure 4.6 Sequence Diagram

4.2.3.5. Component Diagram of the System

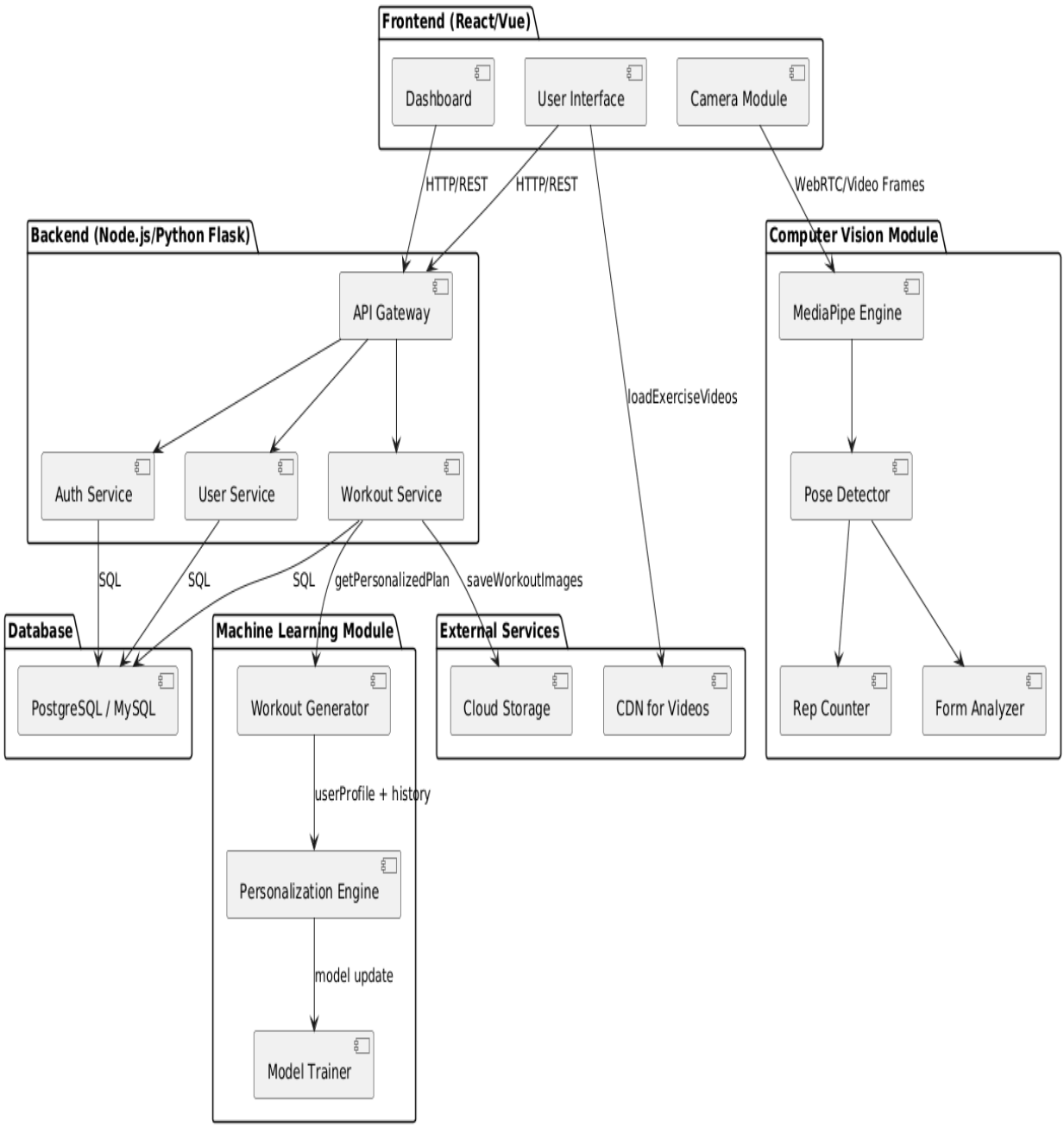


Figure 4.7 Component Diagram

4.3. Chapter Summary

This chapter covered the design of the AI Fitness Coach system. We explained why we chose the Agile software development model and discussed its benefits like flexibility and early delivery, as well as its limitations like the need for constant communication. We described the five-step methodology including data collection, data processing, model development, system design, and implementation. We explained the entity relationship diagram for the database showing how users, workouts, exercises, and sets relate to each other. We discussed the UML diagrams we created including the use case diagram, class diagram, activity diagram, sequence diagram, and component diagram. Each of these diagrams helps us understand a different aspect of the system. The next chapter will cover the database design in more detail including the specific tables and queries we will use.

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